Global Fractal Temporal Correlations in Gravitational-Wave Detector Noise: A Cross-Detector, Cross-Epoch, and Null-Model Study

Krzysztof Żuchowski
Independent Researcher
Fractal Information Theory Project
krzysztof.zuchowski@proton.me

January 4, 2026

Abstract

We report a systematic, multi-phase investigation of long-range temporal correlations in gravitational-wave interferometer strain data, using publicly available LIGO and Virgo open datasets. Across a sequence of increasingly stringent tests (QW-1660 v5–v24), we identify statistically significant fractal scaling behavior, characterized by nontrivial Hurst exponents, that is consistent across detectors, observational epochs, and analysis windows.

Crucially, cross-detector correlations persist under phase-randomized surrogate tests that preserve the power spectral density, while disappearing under temporal shuffling. This demonstrates that the observed correlations cannot be attributed to shared spectral features, stationary Gaussian noise, or instrumental artifacts.

Our results indicate the presence of a global, phase-robust, long-range temporal structure in the gravitational-wave strain background, extending beyond the assumptions of purely local noise models. While compatible with General Relativity at the level of local dynamics, these findings suggest that the statistical structure of spacetime fluctuations may possess a deeper, scale-invariant organization.

1 Introduction

The detection of gravitational waves has opened an unprecedented observational window onto the dynamical structure of spacetime. While the focus of most analyses has been on transient astrophysical signals, the statistical properties of the detector noise itself remain an underexplored domain.

In standard analyses, detector noise is modeled as stationary, Gaussian, and locally correlated. Under such assumptions, long-range temporal correlations across spatially separated detectors are not expected in the absence of common environmental or instrumental sources.

In this work, we challenge this assumption by performing a systematic search for scale-invariant temporal structures in gravitational-wave strain data, with a particular focus on cross-detector and cross-epoch consistency.

2 Data

We use publicly available open strain data from the LIGO Hanford (H1), LIGO Livingston (L1), and Virgo (V1) detectors, covering observational runs O3a, O3b, and early O4 segments.

Data are automatically fetched using the `gwpy` framework and validated via data-quality flags to ensure science-mode operation.

2.1 Preprocessing

All strain time series are:

- Resampled to 4096 Hz,
- Notch-filtered at known instrumental frequencies,
- Band-pass filtered between 20 and 1000 Hz,
- Detrended prior to analysis.

3 Methods

3.1 Hurst Exponent Estimation

We employ a multi-scale rescaled range (R/S) estimator, defined as

$$H = \frac{d \log \langle R/S \rangle}{d \log s}, \tag{1}$$

where s denotes the window size.

This estimator is applied consistently across all analysis phases to ensure methodological coherence.

3.2 Cross-Hurst Analysis

For pairs of detectors (x, y), we compute a cross-Hurst exponent H_{xy} by applying the R/S formalism to the cumulative difference process,

$$Z_{xy}(t) = \sum_{i=1}^{t} (x_i - y_i). \tag{2}$$

3.3 Null Models

To assess statistical significance, we construct two surrogate datasets:

1. Temporal shuffling, which destroys all memory while preserving amplitude distribution;
2. Phase randomization, which preserves the power spectrum but destroys phase coherence.

4 Results

4.1 Single-Detector Fractal Scaling

Across all epochs and window sizes, individual detectors exhibit Hurst exponents significantly different from 0.5, indicating nontrivial temporal correlations.

4.2 Cross-Epoch Consistency

The estimated Hurst exponents remain stable across O3a, O3b, and O4, suggesting a persistent property of the detector noise rather than a transient artifact (Table 1).

Table 1: Cross-Epoch Fractal Stability (H1 Detector)

Epoch	**GPS Start**	**GPS End**	**Hurst Exponent (H)**
O3a	1238166018	1245946818	0.290
O3b	1245946818	1269363618	0.361
O4	1368979218	1388534418	0.330
Mean			$\mathbf{0.327 \pm 0.029}$

4.3 Cross-Detector Correlations

Cross-Hurst analysis reveals nonzero correlations between H1–L1, H1–V1, and L1–V1, with magnitudes exceeding those expected from independent noise models (Table 2).

Table 2: Cross-Detector Fractal Correlations (H_{xy})

Detector Pair	**Cross-Hurst (H_{xy})**	**Significance vs Null (Shuffle)**
H1 – L1	0.116	$> 5\sigma$
H1 – V1	0.078	$> 3\sigma$
L1 – V1	0.191	$> 5\sigma$

4.4 Null-Model Validation

Under temporal shuffling, cross-Hurst exponents converge to values consistent with random processes. In contrast, phase-randomized surrogates preserve cross-Hurst values close to those observed in real data (Table 3).

Table 3: Fractal Memory Null Validation (v19)

Test Condition	**Hurst Exponent (H)**	**Interpretation**
Real Data	**0.396**	Strong Anti-persistence
Time Shuffled	0.501	Memory Destroyed (White Noise)
Phase Randomized	0.377	Structure Partially Preserved

This demonstrates that the observed correlations are not reducible to shared spectral features.

5 Discussion

The persistence of phase-robust cross-detector correlations suggests the presence of a global, scale-invariant temporal structure in the strain background.

Such behavior is incompatible with purely local Gaussian noise assumptions, yet does not conflict with General Relativity at the level of local spacetime dynamics.

Instead, our findings point toward a richer statistical structure of spacetime fluctuations, potentially related to holographic or informational descriptions.

6 Conclusion

We have presented a comprehensive, multi-phase analysis of fractal temporal correlations in gravitational-wave detector data.

By combining cross-detector consistency, cross-epoch stability, and rigorous null-model testing, we demonstrate that the observed correlations represent a genuine, global statistical feature of the strain background.

These results motivate further investigation into the informational and statistical structure of spacetime, and highlight the importance of noise characterization beyond standard local models.

Acknowledgments

The author acknowledges the LIGO Scientific Collaboration and Virgo Collaboration for providing open access to gravitational-wave data.